

Prosit 1: JobBoard case study: architectural diagnosis

Équipe de 4 à 5 personnes · 1 heures

A. Learning objectives:

At the end of this prosit, the student will be able to:

- Discover the functional context of the **JobBoard** application.
- Analyze an existing application developed according to a monolithic architecture.
- Identify the main functionalities of the application.
- Recognize the characteristics of tightly coupled code (spaghetti programming).
- Identify the limitations of a monolithic architecture in the face of evolving needs.
- Analyze the difficulties related to maintenance, scalability and deployment of a monolithic application.
- Identify the actors and the main business processes of the application.
- Propose architectural improvement avenues by justifying the needs for an evolution toward a distributed architecture.

B. Pedagogical method:

TEAM	DURATION	MODALITY
3 to 5 students	2 hours	PBL — no code

C. Required work:

1. The situation

TalentConnect is a Tunisian SME (35 people) that publishes **JobBoard**, an online employment platform. The application is developed in **classical Symfony MVC** (180,000 lines of code, MySQL, 12 developers). It currently serves **800,000 users** in 6 countries.

At the end of 2025, management signs 3 strategic contracts: **expansion to 3 new countries (× 4 traffic)**, **ML recommendation engine (Python)**, **B2B APIs for partners**. In parallel, the teams report: 2 h 30 min deployments that fail, saturated search engine, Git conflicts, 3 h outage in March 2026 (€40k loss), refusal to migrate to Symfony 6.4.

What the CEO is asking for

*“Our application works, but we feel it will not hold up under the load of the coming months. We want to understand what is blocking and what our options are. **No solution yet: a clear and honest diagnosis.**”*

2. What is expected of you

At the end of this Prosit, you will be able to: **map an existing application, identify the limitations of a monolith, analyze non-functional requirements, conduct a literature review and use AI critically** — to argue your recommendations before a jury.

Important

This Prosit requires **no line of code**. You must **not** propose a definitive technical solution. You are **consultants who analyze** and open up avenues.

3. The work to be done — 7 sequential stages

Each stage builds on the previous ones. Follow them in order.

- | | |
|---|--|
| 1 | Analyze the existing application
Identify the actors, the functionalities and the business processes. Produce a functional map in the form of a diagram (mind map). |
| 2 | Study the current MVC architecture
Recall the MVC pattern. Schematically represent the Symfony architecture (layers, database, deployment). Identify what is shared among all modules. |
| 3 | Identify the limitations of the monolith
For each warning signal observed at JobBoard, identify the architectural cause. Classify the limitations (technical, organizational, economic). |
| 4 | Analyze the new needs
For each strategic contract, list the functional and non-functional requirements. Can they be satisfied by the current architecture? |
| 5 | Conduct a literature review
Consult at least 3 academic sources (Fowler, Newman, Coulouris, Sommerville, IEEE, ACM). Produce a 2 to 3 page synthesis. |
| 6 | Confront your analysis with an AI
Query at least 2 AIs (ChatGPT, Claude, Gemini...) with precise prompts. Analyze the responses critically. Verify them. |

7

Prepare the presentation

Consolidate the final report and the oral presentation (15 min + 10 min Q&A) before the teacher and the other teams.

4. Critical use of AI

AI is your **analysis assistant**, never your replacement. Here are the rules:

1

You have first produced your own analysis (stages 1 to 5).

2

AI is used to challenge, enrich, question — never to replace.

3

Every interaction is documented: tool, date, prompt, response.

4

Every key claim is verified by an academic source.

5

The use of AI is declared in the final report.

5. Your deliverables

1

D1**Architectural analysis report (15 to 25 pages, PDF)**

Structured according to the 7 stages: mapping, current architecture, identified limitations, requirements of the new contracts, documentary synthesis, preliminary recommendations, open conclusion.

2

D2**Mapping & architecture diagrams**

Diagrams: functional diagram of domains, current layered architecture, actors and interactions of the system. Integrated into the report.

3

D3**Critical appendix on the use of AI (5 to 10 pages)**

All the prompts, the responses, your critical analysis (5 axes), the verification table by academic sources, a final reflection.

4	D4	Presentation support (10 to 15 slides, PPTX/PDF) Professional slides with cited references. A final slide summarizing the open questions that remain.
5	D5	Oral presentation (15 min + 10 min of questions) Before the teacher and the other teams playing the role of the steering committee. Equitable distribution of speaking time.

Learning objectives covered through Prosit 1:

- **LO1:** Explain the evolution of software architectures up to modern distributed architectures.
- **LO2:** Compare the Monolithic, SOA and Microservices architectures according to different criteria (coupling, deployment, evolvability, maintenance, scalability).
- **LO3:** Analyze the business needs leading to the adoption of a distributed architecture.